

Problem D. ROT N

Time limit 2000 ms

Mem limit 1048576 kB

Problem Statement

We have a string S consisting of uppercase English letters. Additionally, an integer N will be given.

Shift each character of S by N in alphabetical order (see below), and print the resulting string.

We assume that **A** follows **Z**. For example, shifting **A** by 2 results in **C** (**A** $\rightarrow$ **B** $\rightarrow$ **C**), and shifting **Y** by 3 results in **B** (**Y** $\rightarrow$ **Z** $\rightarrow$ **A** $\rightarrow$ **B**).

Constraints

- $0 \leq N \leq 26$
- $1 \leq |S| \leq 10^4$
- S consists of uppercase English letters.

Input

Input is given from Standard Input in the following format:

```
N
S
```

Output

Print the string resulting from shifting each character of S by N in alphabetical order.

Sample 1

Input	Output
2 ABCXYZ	CDEZAB

Note that **A** follows **Z**.

Sample 2

Input	Output
0 ABCXYZ	ABCXYZ

Sample 3

Input	Output
13 ABCDEFGHIJKLMNOPQRSTUVWXYZ	NOPQRSTUVWXYZABCDEFGHIJKLM